

Git Commands Handbook

Corporate Training Edition - Beginner to Intermediate

1. Git Basics

Command	Purpose	Example
git --version	Check Git version	git --version
git config --global user.name	Configure username	git config --global user.name "Swapna"
git config --global user.email	Configure email	git config --global user.email "user@example.com"
git config --list	View configuration	git config --list

2. Repository

Command	Purpose	Example
git init	Initialize repository	git init
git clone URL	Clone repository	git clone https://github.com/user/repo.git
git status	Repository status	git status

3. Staging

Command	Purpose	Example
git add file	Stage file	git add app.java
git add .	Stage all	git add .
git restore --staged file	Unstage file	git restore --staged app.java

4. Commits

Command	Purpose	Example
git commit -m	Commit changes	git commit -m "Initial commit"
git log	Commit history	git log
git log --oneline	Compact history	git log --oneline

5. Branching

Command	Purpose	Example
git branch	List branches	git branch
git branch feature	Create branch	git branch feature
git switch feature	Switch branch	git switch feature
git switch -c feature	Create and switch	git switch -c feature

git merge feature	Merge branch	git merge feature
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6. Remote

Command	Purpose	Example
git remote -v	View remotes	git remote -v
git remote add origin URL	Add remote	git remote add origin https://github.com/user/repo.
git push -u origin main	First push	git push -u origin main
git push	Push commits	git push
git pull	Fetch and merge	git pull
git fetch	Download changes	git fetch

7. Undo & Recovery

Command	Purpose	Example
git restore file	Discard working changes	git restore app.java
git reset --soft HEAD~1	Undo commit keep staged	git reset --soft HEAD~1
git reset --hard HEAD	Discard all local changes	git reset --hard HEAD
git revert HASH	Reverse commit	git revert abc1234

8. Stash & Tags

Command	Purpose	Example
git stash	Save work	git stash
git stash pop	Restore stash	git stash pop
git tag v1.0	Create tag	git tag v1.0
git push origin v1.0	Push tag	git push origin v1.0

9. Inspection

Command	Purpose	Example
git diff	Working changes	git diff
git diff --staged	Staged changes	git diff --staged
git show HASH	Commit details	git show HEAD

Typical Git Workflow

git init → git status → git add . → git commit -m "message" → git branch feature → git switch feature → edit files → git add . → git commit -m "feature" → git switch main → git merge feature → git remote add origin URL → git push -u origin main

Practice Exercises

1. Initialize a new repository.
2. Create README.md and commit it.
3. Create a feature branch and switch to it.
4. Modify a file and inspect differences.
5. Merge the feature branch into main.
6. Push the repository to GitHub.
7. Use git stash before switching branches.
8. Recover from an accidental commit using git reset or git revert.